Numerical Analysis of the Black–Scholes PDE: Stability and Convergence of Finite Difference Schemes

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Abstract

Options in finance are derivative contracts that give investors the right, but not the obligation, to buy (call) or sell (put) an underlying asset. The Black-Scholes partial differential equation is one of the most important equations in financial mathematics and is widely used in the pricing of European-style options. Although a closed-form analytical solution exists in the classical setting (constant volatility, exercise at maturity), numerical methods are essential for more general pricing problems and are also useful for understanding the behavior of discretization schemes. In this paper, the Black-Scholes PDE is transformed into the standard heat equation through a logarithmic coordinate transformation, a reversal of time, and an exponential scaling of the solution. The full transformation is derived in detail, including the computation of the constants α and β and the transformed boundary conditions for the new dependent variable u(x,τ). Three finite difference schemes are then studied: Forward Time Centered Space (FTCS), Backward Time Centered Space (BTCS), and the Crank–Nicolson method. Their stability, consistency, convergence, and numerical behavior are analyzed theoretically and illustrated computationally. The results show that FTCS is easy to implement but conditionally stable, BTCS is unconditionally stable but more diffusive, and Crank–Nicolson offers the best balance of stability and accuracy among the methods considered.

1. Introduction

The Black–Scholes equation is a classical partial differential equation that arises in the pricing of derivatives in financial mathematics. For a European call option with strike price K, maturity T, volatility σ, and risk-free interest rate r, the option value V(S,t) satisfies

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS\frac{\partial V}{\partial S} - rV = 0, \quad V = V(S,t),\ S > 0,\ 0 \le t < T$$

The terminal condition for a European call option is

$$V(S,T) = \max(S - K, 0)$$

The Black-Scholes model is fundamental because it provides a mathematically elegant framework for option pricing and has historically played a major role in the development of

modern financial engineering. However, exact formulas are available only under restrictive assumptions such as constant volatility. From the viewpoint of scientific computing, numerical methods are important both because they provide approximations when closed-form solutions are unavailable and because they allow direct investigation of numerical properties such as consistency, stability, and convergence.

The main goals of this project are:

- Transform the Black-Scholes PDE into the classical heat equation.
- Derive finite difference discretization for the transformed PDE.
- Compare FTCS, BTCS, and Crank-Nicolson schemes.
- Analyze their stability and convergence behavior.
- Compare numerical solutions with the analytical Black–Scholes formula.

2. Financial and Mathematical Interpretation

The Black–Scholes PDE contains several important terms:

- $\frac{\partial V}{\partial t}$: time sensitivity of the option value.
- $\frac{\partial V}{\partial S}$: first-order sensitivity to the asset price.
- $\frac{\partial^2 V}{\partial S^2}$: curvature of the option value with respect to the asset price.
- $\frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2}$: Diffusion term
- $rS\frac{\partial V}{\partial S}$: drift-type contribution.
- $-rV$: discounting term (to get present value of the option).

From a scientific computing perspective, the variable coefficient S^2 in front of $\frac{\partial^2 V}{\partial S^2}$ is what makes direct discretization awkward, it means the diffusion coefficient varies across the grid. The logarithmic transformation in Section 4 eliminates this by converting to a constant-coefficient PDE, which is exactly why the transformation is useful numerically and not just mathematically.

3. Initial and Boundary Conditions

The terminal condition at maturity is V(S,T) = max(S − K,0). For a European call option, the behavior at the boundaries is

$$V(0,t) = 0$$

$$V(S,t) \approx S - Ke^{(-r(T-t))}, \ S \to \infty$$

The condition V(0,t) = 0 follows from the fact that if the underlying asset price is zero, the right to buy it for a positive strike price has no value. The asymptotic condition for large S comes from the fact that a deep in-the-money call (profitable exercise of option) behaves approximately like the underlying asset minus the discounted strike price.

4. Transformation to the Heat Equation

A standard approach is to transform the Black-Scholes PDE into the one-dimensional heat equation. This is useful because the heat equation has constant coefficients and its numerical behavior is easier to analyze.

4.1 Change of Variables

Define the new variables

$$x = \ln\left(\frac{S}{K}\right), \qquad \tau = \left(\frac{\sigma^2}{2}\right)(T - t)$$

Here, x is a logarithmic measure of the asset price relative to the strike price, and τ reverses time so that τ = 0 corresponds to maturity t = T.

Next, we define a transformed dependent variable u(x,τ) by

$$V(S,t) = Ke^{\alpha x + \beta\tau}u(x,\tau),$$

where α and β are constants to be determined.

4.2 Derivative Calculations

Since $S = Ke^{x}, x = \ln\left(\frac{S}{K}\right)$, and $\tau = \left(\frac{\sigma^2}{2}\right)(T - t)$, we first compute the derivatives of V. Using the chain rule,

$$\frac{\partial\tau}{\partial t} = -\frac{\sigma^2}{2}, \quad \frac{\partial x}{\partial t} = 0, \quad \frac{\partial x}{\partial S} = \frac{1}{S}$$

Therefore,

$$\frac{\partial V}{\partial t} = -\left(\frac{\sigma^2}{2}\right)K\, e^{\alpha x + \beta\tau}\left(\frac{\partial u}{\partial \tau} + \beta u\right)$$

$$\frac{\partial V}{\partial S} = \left(K\frac{e^{\alpha x + \beta\tau}}{S}\right)\left(\frac{\partial u}{\partial x} + \alpha u\right)$$

$$\frac{\partial^2 V}{\partial S^2} = \left(K\frac{e^{\alpha x + \beta\tau}}{S^2}\right)\left[\frac{\partial^2 u}{\partial x^2} + \frac{(2\alpha - 1)\partial u}{\partial x} + \alpha(\alpha - 1)u\right]$$

4.3 Substitution into the Black-Scholes PDE

Substituting these expressions into the Black–Scholes PDE and factoring out $Ke^{\alpha x+\beta\tau}$, we get

$$u_\tau = u_{xx} + \left(2\alpha - 1 + \frac{2r}{\sigma^2}\right)u_x + \left(\alpha^2 - \alpha + \left(\frac{2r}{\sigma^2}\right)\alpha - \frac{2r}{\sigma^2} - \beta\right)u$$

4.4 Computation of α and β

To reduce the transformed PDE to the heat equation, we choose α and β so that the coefficients of u_x and u are eliminated.

First, we set the coefficient of u_x equal to zero:

$$2\alpha - 1 + \frac{2r}{\sigma^2} = 0$$

Hence,

$$\alpha = \frac{1}{2} - \frac{r}{\sigma^2}$$

Now, we set the coefficient of u equal to zero:

$$\alpha^2 - \alpha + \left(\frac{2r}{\sigma^2}\right)\alpha - \frac{2r}{\sigma^2} - \beta = 0$$

So,

$$\beta = \alpha^2 - \alpha + \left(\frac{2r}{\sigma^2}\right)\alpha - \frac{2r}{\sigma^2}$$

After simplification,

$$\beta = -\left(\frac{1}{4}\right)\left(1 + \frac{2r}{\sigma^2}\right)^2$$

Therefore, the transformed PDE becomes

$$\frac{\partial u}{\partial \tau} = \frac{\partial^2 u}{\partial x^2}$$

which is the standard one-dimensional heat equation.

5. Transformed Initial and Boundary Conditions

5.1 Initial Condition for u

At maturity t = T, we have τ = 0. Using the payoff condition and the transformation,

$$K\, e^{\alpha x} u(x,0) = \max(Ke^{x} - K, 0)$$

Dividing by $Ke^{\alpha x}$ gives

$$u(x,0) = e^{-\alpha x} \max(e^{x} - 1,0)$$

5.2 Left Boundary Condition

As S → 0^+, we have x → −∞, and since V(0,t) = 0, it follows that

$$u(x_{min}, \tau) = 0$$

5.3 Right Boundary Condition

As S → ∞, we use the asymptotic condition $V(S,t) \approx S - Ke^{-r(T-t)}$. Since $S = Ke^{x}$ and

$T - t = \frac{2\tau}{\sigma^2}$, we get

$$u(x_{max}, \tau) \approx e^{(1-\alpha)x_{max} - \beta\tau}$$

6. Domain Truncation and Grid Construction

The transformed problem is posed on an unbounded domain in x, but numerical computation requires truncation:

$$x \in [x_{min}, x_{max}], \tau \in [0, \tau_{max}],$$

$$x_{min} = \ln\left(\frac{S_{min}}{K}\right), \quad x_{max} = \ln\left(\frac{S_{max}}{K}\right)$$

$$\tau_{max} = \left(\frac{\sigma^2}{2}\right) T$$

For the scope of this project, I'll be using $S_{min} \approx 0$ & $S_{max} \approx 10K$

We introduce the grid

$$x_j = j\Delta x, \quad j = 0,1,\dots,N$$

$$\tau_n = n\Delta t, \quad n = 0,1,\dots,M$$

Let $u_j^n \approx u(x_j, \tau_n)$, we define

$$\lambda = \frac{\Delta t}{(\Delta x)^2}$$

7. Finite Difference Discretization

The heat equation can be discretized using standard finite difference approximations:

$$u_{\tau(x_j,\tau_n)} \approx \frac{u_j^{n+1} - u_j^n}{\Delta t}$$

$$u_{xx(x_j,\tau_n)} \approx \frac{u_{j+1}^n - 2u_j^n + u_{j-1}^n}{(\Delta x)^2}$$

7.1 FTCS Scheme

The Forward Time Centered Space scheme uses a forward difference in time and centered difference in space:

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} = \frac{u_{j+1}^n - 2u_j^n + u_{j-1}^n}{(\Delta x)^2}$$

Hence,

$$u_j^{n+1} = u_j^n + \lambda\left(u_{j+1}^n - 2u_j^n + u_{j-1}^n\right)$$

This scheme is explicit and easy to implement.

7.2 BTCS Scheme

The Backward Time Centered Space scheme uses a backward difference in time and centered difference in space at the new time level:

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} = \frac{u_{j+1}^{n+1} - 2u_j^{n+1} + u_{j-1}^{n+1}}{(\Delta x)^2}$$

Rearranging gives

$$-\lambda u_{j-1}^{n+1} + (1+2\lambda)u_j^{n+1} - \lambda u_{j+1}^{n+1} = u_j^n.$$

This scheme is implicit and requires solving a tridiagonal linear system at each time step.

7.3 Crank–Nicolson Scheme

The Crank–Nicolson method averages the spatial second derivative between time levels n and n+1:

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} = \left(\frac{1}{2}\right)[u_{xx}^n + u_{xx}^{n+1}]$$

After discretization and rearrangement,

$$-\left(\frac{\lambda}{2}\right)u_{j-1}^{n+1} + (1+\lambda)u_j^{n+1} - \left(\frac{\lambda}{2}\right)u_{j+1}^{n+1} = \left(\frac{\lambda}{2}\right)u_{j-1}^n + (1-\lambda)u_j^n + \left(\frac{\lambda}{2}\right)u_{j+1}^n.$$

This method is implicit and more accurate than BTCS as it is second-order in time.

8. Stability Analysis

8.1 Von Neumann Analysis for FTCS

To analyze stability of FTCS, assume a Fourier mode of the form

$$u_j^n = G^n e^{ikj\Delta x}$$

where G is the amplification factor.

Substituting into the FTCS scheme gives

$$G = 1 - 4\lambda \sin^2\left(\frac{k\Delta x}{2}\right)$$

For stability, we require $|G| \leq 1$. Since the maximum of $\sin^2$ is 1, the most restrictive case gives

$$-1 \leq 1 - 4\lambda \leq 1$$

which gives

$$\lambda \leq \frac{1}{2}$$

Thus, FTCS is conditionally stable.

8.2 BTCS Stability

BTCS is unconditionally stable. Although it requires solving a linear system at each time step, no restriction is imposed on Δt relative to Δx.

8.3 Crank–Nicolson Stability

Crank–Nicolson is also unconditionally stable for the heat equation. In addition, it is second-order accurate in time. It generally preserves accuracy better than BTCS.

9. Consistency and Convergence

A finite difference scheme is consistent if its truncation error tends to zero as $\Delta x, \Delta t \to 0$. A scheme is convergent if the numerical solution approaches the true solution of the PDE as the grid is refined. i.e.,

$$\lim_{\Delta x, \Delta t \to 0} u_j^n = u(x,t)$$

9.1 Truncation Error

For FTCS, the local truncation error is

$$O(\Delta t) + O(\Delta x^2),$$

so, FTCS is first-order accurate in time and second-order accurate in space.

For BTCS, the local truncation error is also

$$O(\Delta t) + O(\Delta x^2).$$

For Crank–Nicolson, the local truncation error is

$$O(\Delta t^2) + O(\Delta x^2),$$

so, Crank–Nicolson is second-order in time and second-order in space.

9.2 Lax Equivalence Principle

For linear initial value problems, the guiding principle is

$$\text{Consistency} + \text{Stability} \Rightarrow \text{Convergence}.$$

Thus:

- FTCS converges only when the stability restriction $\lambda \leq 1/2$ is satisfied.
- BTCS converges without a time-step restriction.
- Crank–Nicolson converges and generally gives better accuracy.

10. Numerical Behavior and Interpretation

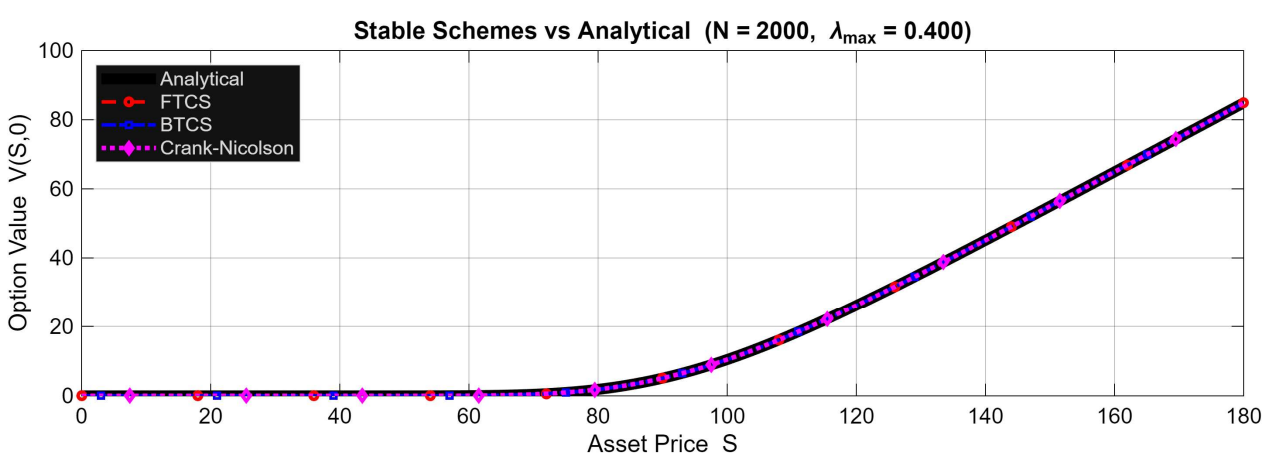

Figure 1. Stable comparison of FTCS, BTCS, and Crank–Nicolson with the analytical solution.

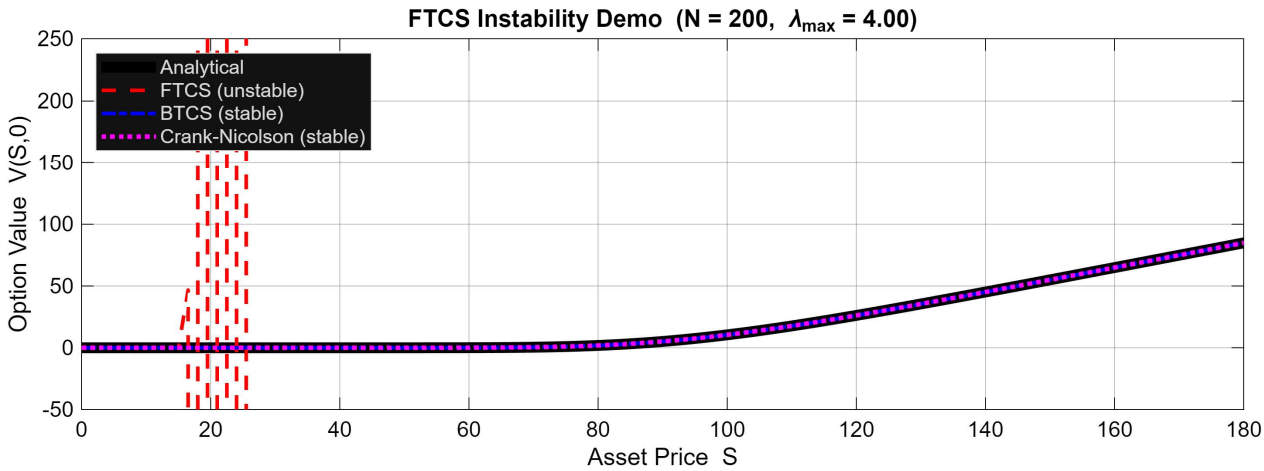

Figure 2. Unstable FTCS behavior when the stability condition is violated. Large oscillations develop, while BTCS and Crank–Nicolson remain stable.

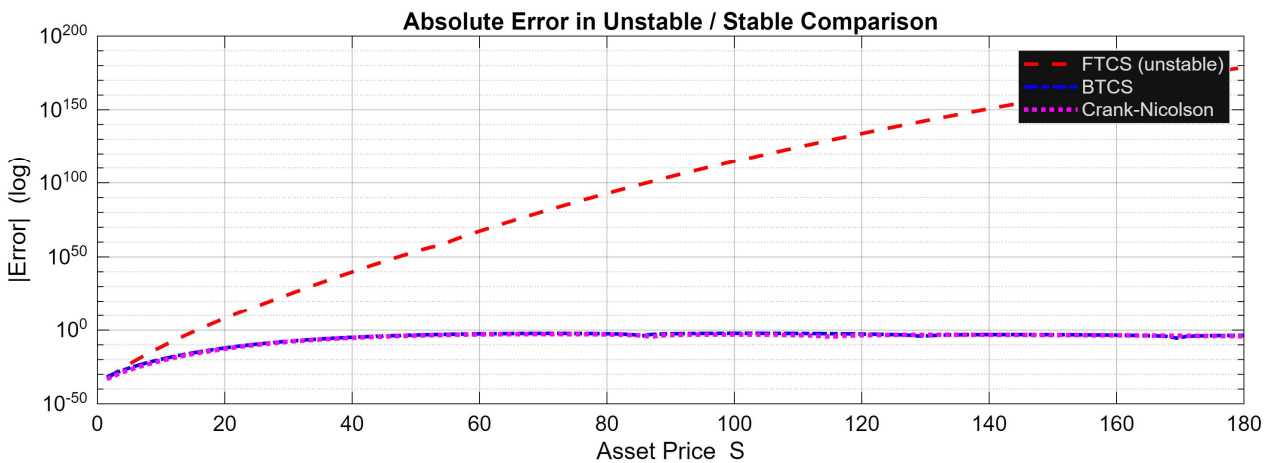

Figure 3. Absolute error (log scale) versus asset price for the unstable FTCS scheme compared with BTCS and Crank–Nicolson. When the stability condition is violated, the FTCS solution exhibits rapidly growing errors and divergence, while BTCS and Crank–Nicolson remain stable and maintain low error across the domain

The numerical results of this project confirm the theoretical analysis.

In the stable case, all three schemes approximate the analytical option price well. The FTCS method also performs well when the stability condition is respected.

In the unstable case, FTCS produces oscillations and eventually diverges. This behavior is expected from the stability analysis.

From a computational point of view:

- FTCS is the simplest and cheapest per time step.
- BTCS is more robust but requires a linear solve.
- Crank–Nicolson combines robustness with higher accuracy.

11. Scheme Comparison

Scheme	Time Accuracy	Stability	Remarks
FTCS	First order	Conditional	Easy to implement, low cost, unstable if $\lambda > 1/2$
BTCS	First order	Unconditional	Stable for all step sizes, but more diffusive
Crank–Nicolson	Second order	Unconditional	Best balance of accuracy and stability

12. Conclusion

This paper studied the numerical solution of the Black-Scholes PDE using finite difference methods. By applying a logarithmic transformation in space, reversing time, and scaling the dependent variable exponentially, the Black-Scholes equation was reduced to the standard heat equation. The constants in the transformation were computed explicitly as

$$\alpha = \frac{1}{2} - \frac{r}{\sigma^2}, \qquad \beta = -\left(\frac{1}{4}\right)\left(1 + \frac{2r}{\sigma^2}\right)^2$$

The transformed initial and boundary conditions for $u(x,\tau)$ were derived, allowing the problem to be solved numerically on a finite computational domain.

Three schemes were considered: FTCS, BTCS, and Crank-Nicolson. Theoretical analysis showed that FTCS is only conditionally stable, while BTCS and Crank-Nicolson are unconditionally stable. Numerical results confirmed these conclusions. FTCS performs well only when its stability condition is respected; otherwise it becomes oscillatory and diverges. BTCS is robust but somewhat diffusive. Crank-Nicolson offers the best overall performance due to its unconditional stability and higher order of temporal accuracy.

Overall, this project demonstrates how ideas from partial differential equations, numerical analysis, and scientific computing can be combined to study a fundamental model in quantitative finance. It also shows that scheme selection matters greatly: stability alone is not sufficient, and the best method is the one that balances stability, efficiency, and accuracy.

References

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